

Setup and Backup of Media Wiki on LAMP (Ubuntu) / WAMP (Windows)



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Implemented by-
Vijaylaxmi Ittannavar

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1. Introduction

MediaWiki is an open-source wiki software platform used to power collaborative websites such as Wikipedia. It allows multiple users to create, edit, and manage pages through a browser-based interface, making it a powerful tool for knowledge sharing in organizations, educational institutions, and public communities.

The purpose of this project is to set up and configure MediaWiki on a server environment, ensuring it is accessible through a web browser and fully operational. The installation will be attempted first on an Ubuntu-based LAMP stack within a virtual machine. If technical limitations prevent a successful LAMP setup, the installation will be carried out on a Windows-based WAMP stack.

Beyond deployment, this project also focuses on implementing a reliable backup mechanism to ensure that all MediaWiki data can be restored in the event of data loss or system failure.

2. Objective

The primary objectives of this project are:

1. Deploy MediaWiki
 - Set up a fully functional MediaWiki installation on a LAMP (Linux, Apache, MySQL, PHP) environment in an Ubuntu virtual machine.
 - In case of critical technical issues, deploy MediaWiki on a WAMP (Windows, Apache, MySQL, PHP) environment as a fallback.
2. Document the Process
 - Record each step taken during installation and configuration.
 - Include details of any problems encountered and the troubleshooting measures applied.
3. Implement a Backup Solution
 - Create a mechanism to take daily backups of MediaWiki data, including:
 - MySQL database.
 - MediaWiki application files and uploaded content.
 - Test the restoration process to confirm data integrity and recovery capability.

The successful completion of this project will result in a fully operational MediaWiki site with a proven disaster recovery plan, and a professional document detailing the complete setup journey.

3. Tools and Environment

- **Operating System:** Ubuntu (LAMP) or Windows (WAMP)
- **Web Server:** Apache
- **Database:** MySQL / MariaDB
- **Scripting Language:** PHP
- **Virtualization Software:** VirtualBox or VMware
- **MediaWiki Version:** (Specify version)
- **Backup Tools:** mysqldump, tar (for LAMP) / WAMP backup utilities
- **Browser:** Firefox / Chrome

4. Prerequisites

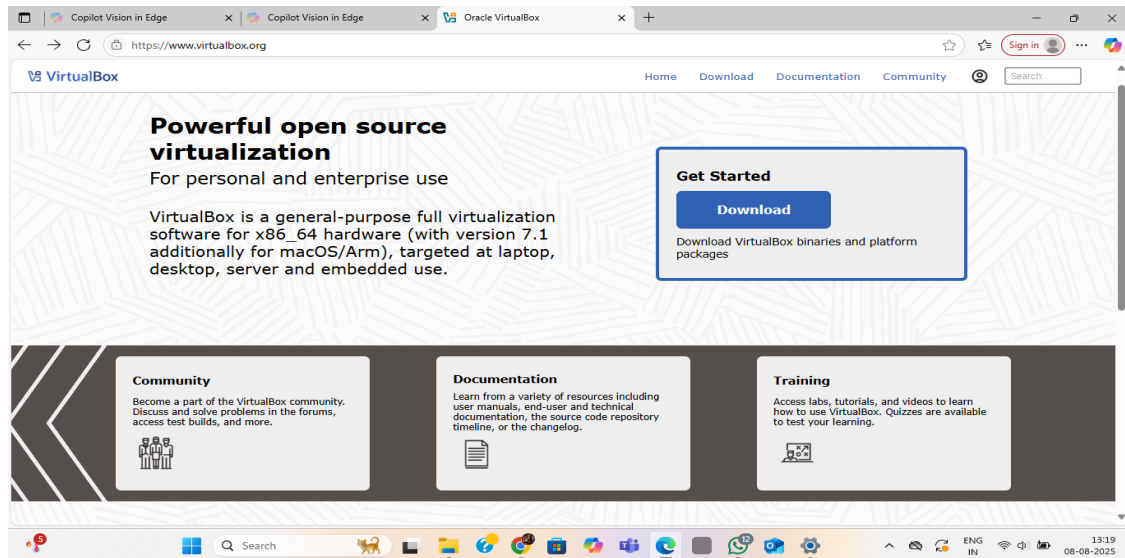
- Minimum server requirements (CPU, RAM, disk space)
- Internet access for downloading packages
- Administrative privileges (sudo in Linux, Administrator in Windows)

5. STEP-BY-STEP GUIDE

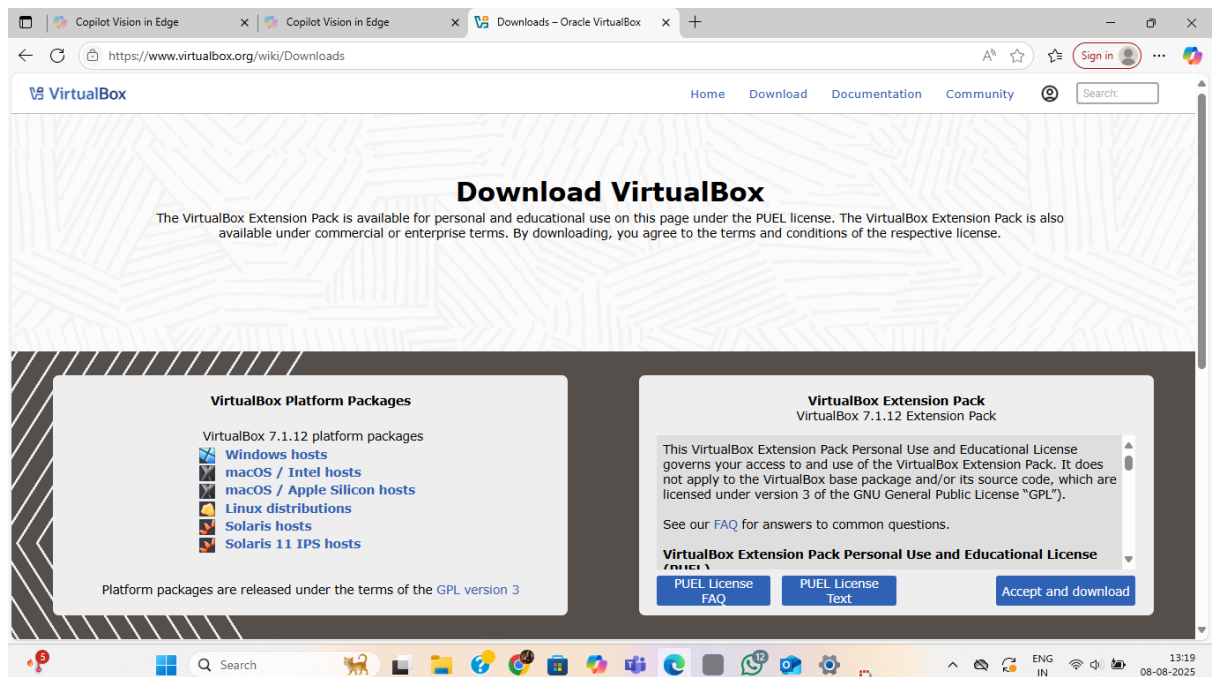
5.1: Create a Virtual Machine (VM) Using VirtualBox

Install VirtualBox

1. • Go to <https://www.virtualbox.org>
2. • Download and install it for your OS (Windows/Mac/Linux)

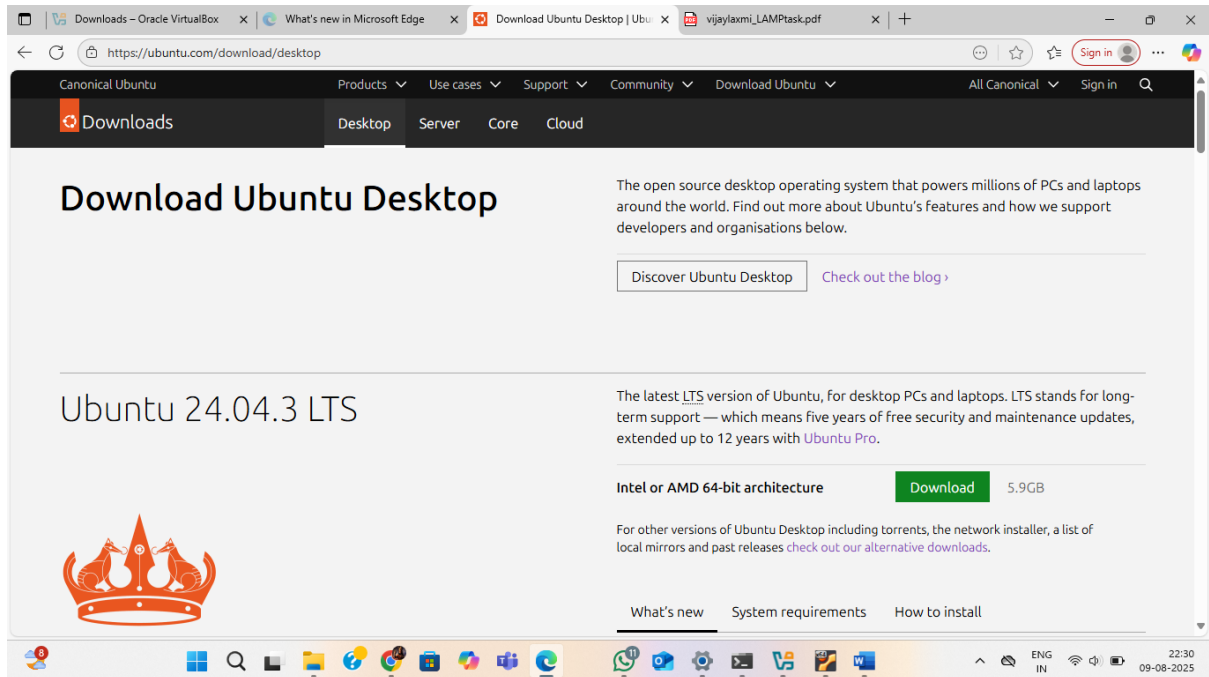


Choose that windows host

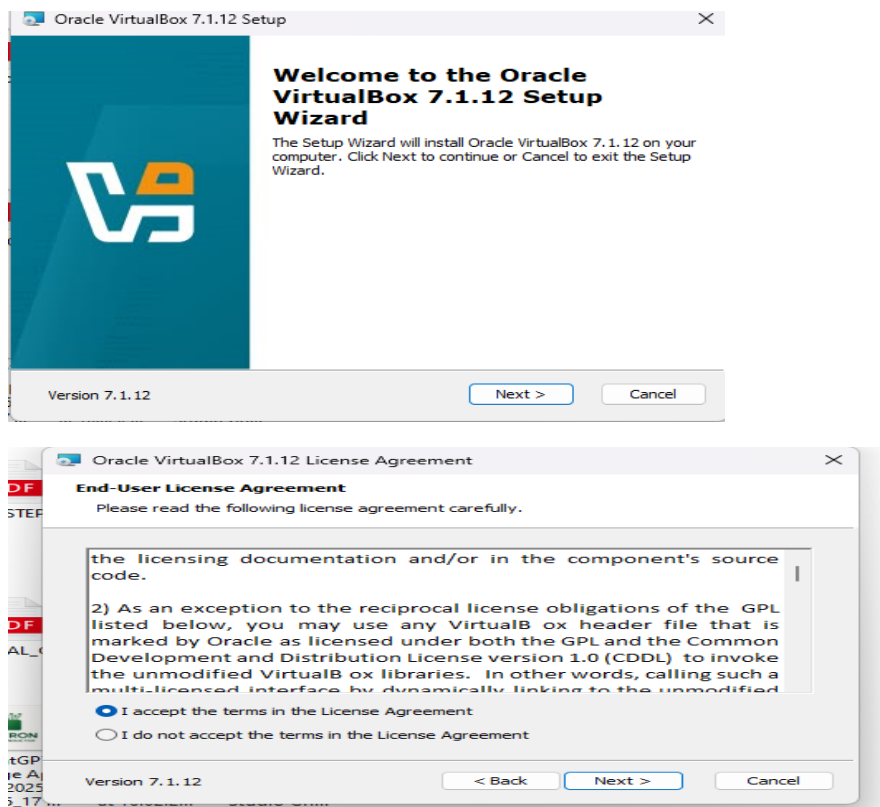


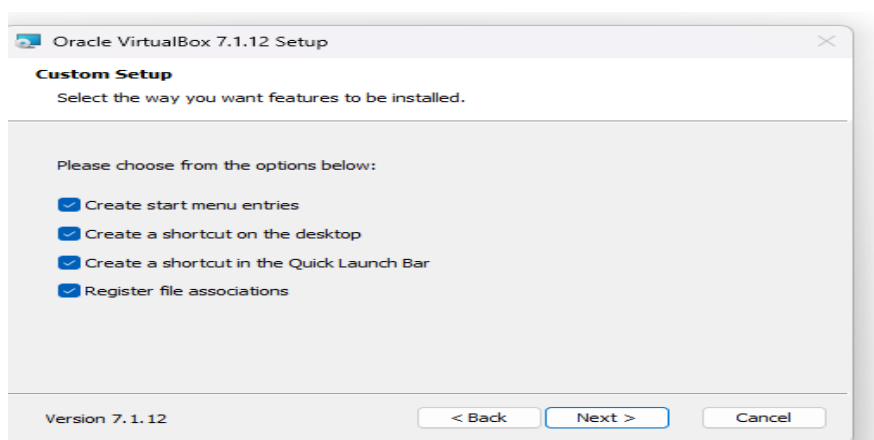
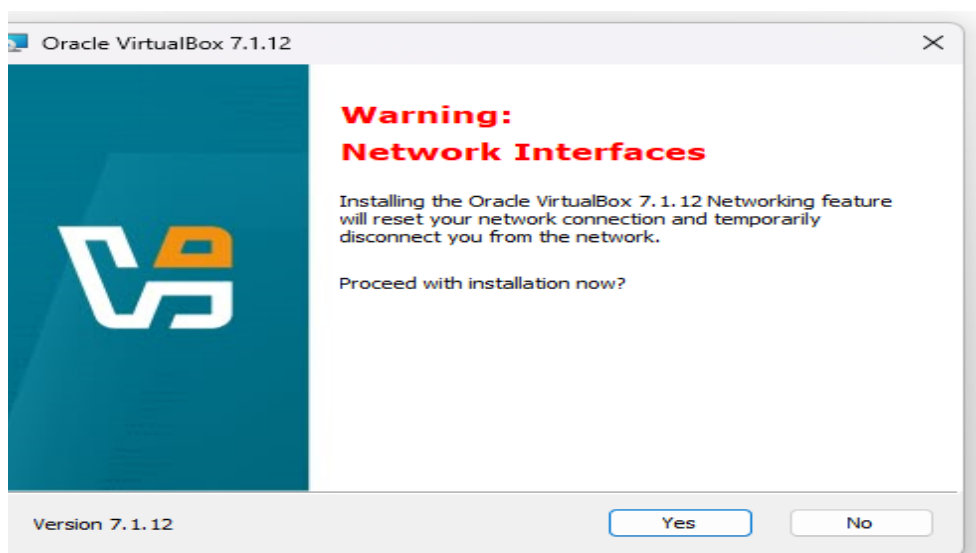
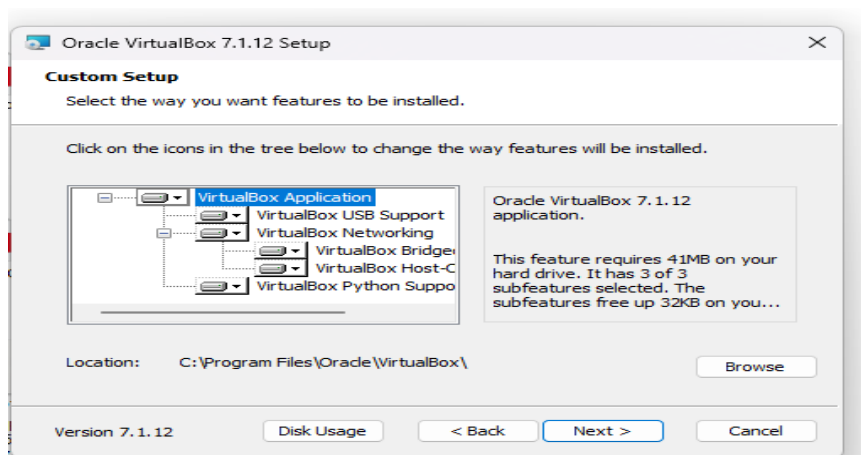
2) Download Ubuntu ISO

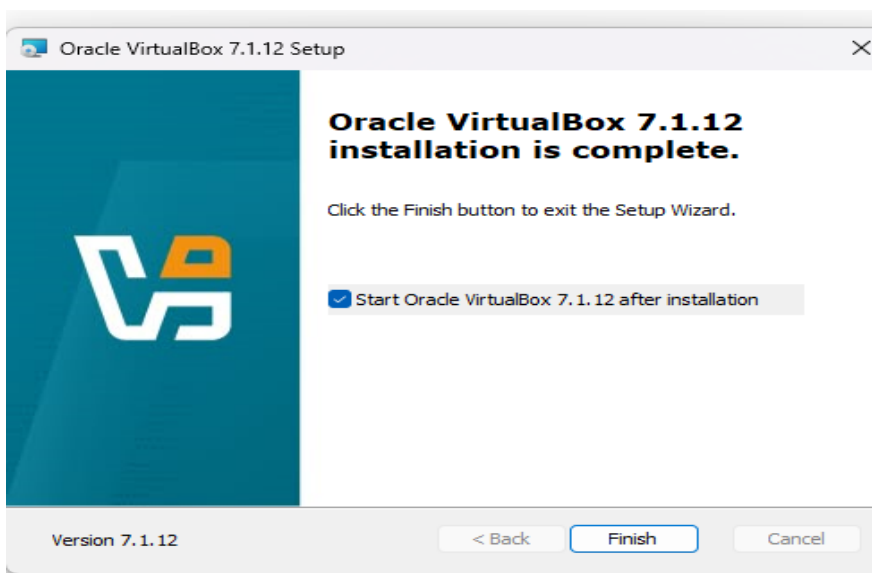
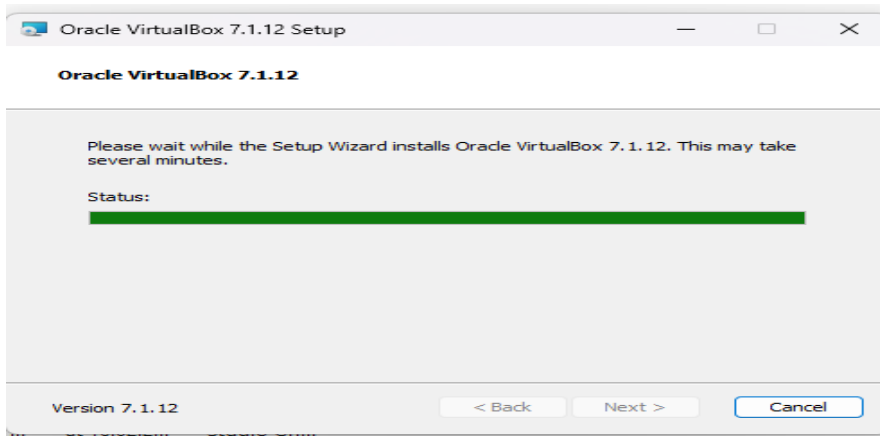
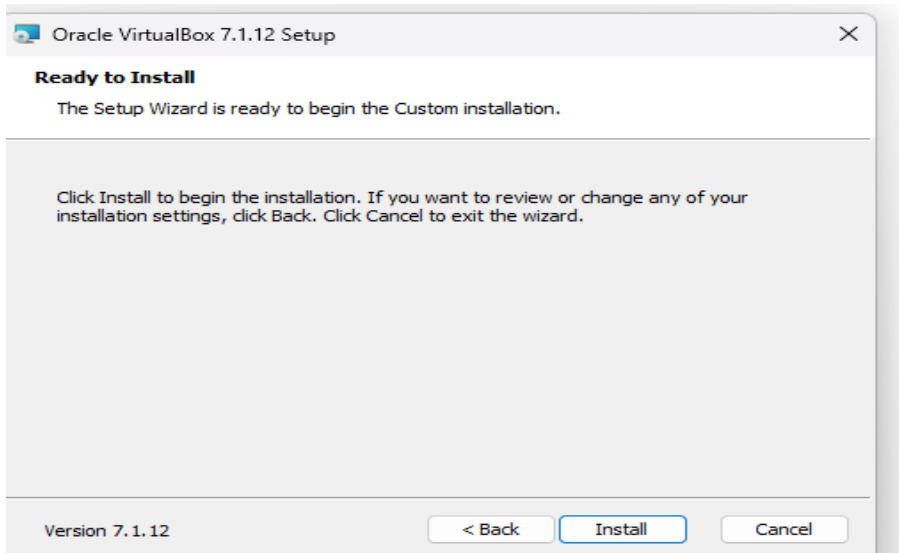
- Go to <https://ubuntu.com/download/desktop>
- Download latest Ubuntu Desktop ISO (around 4 GB)



3. Create a VM in VirtualBox



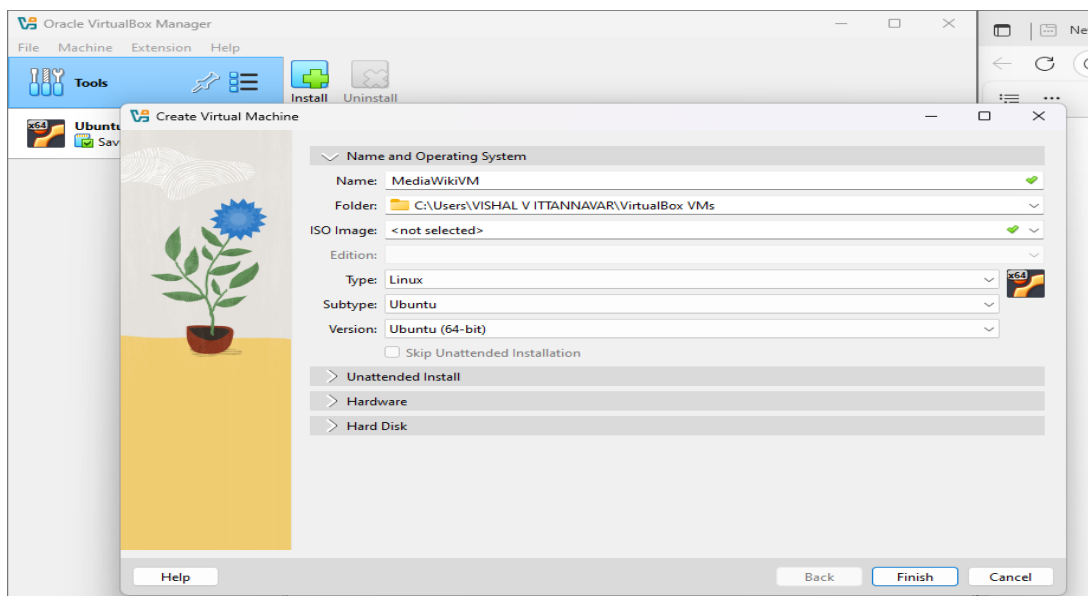
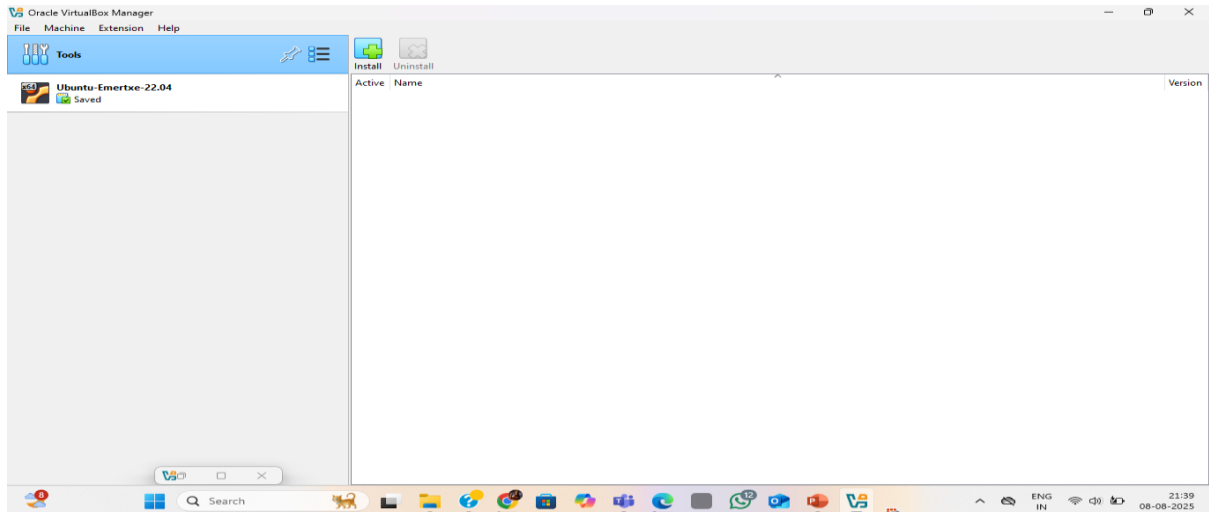


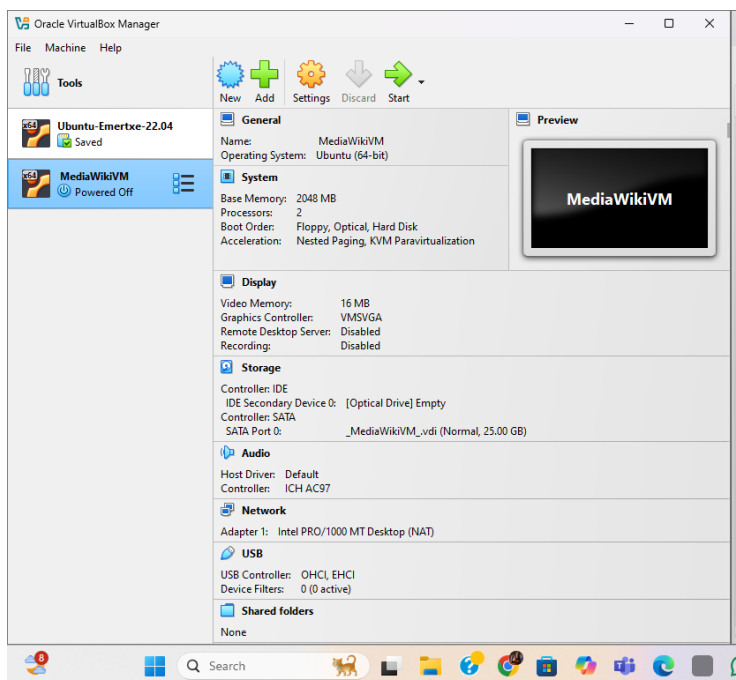
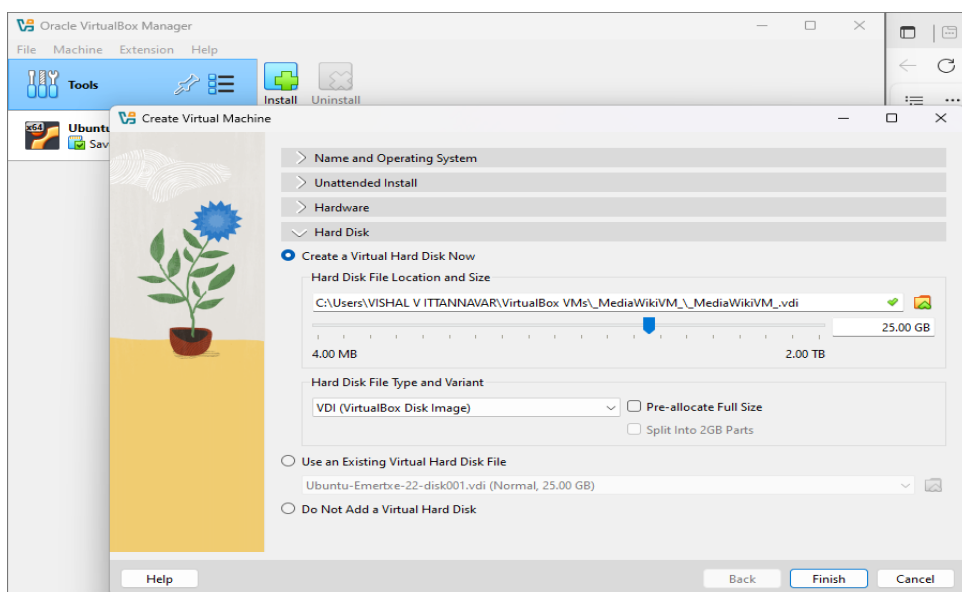
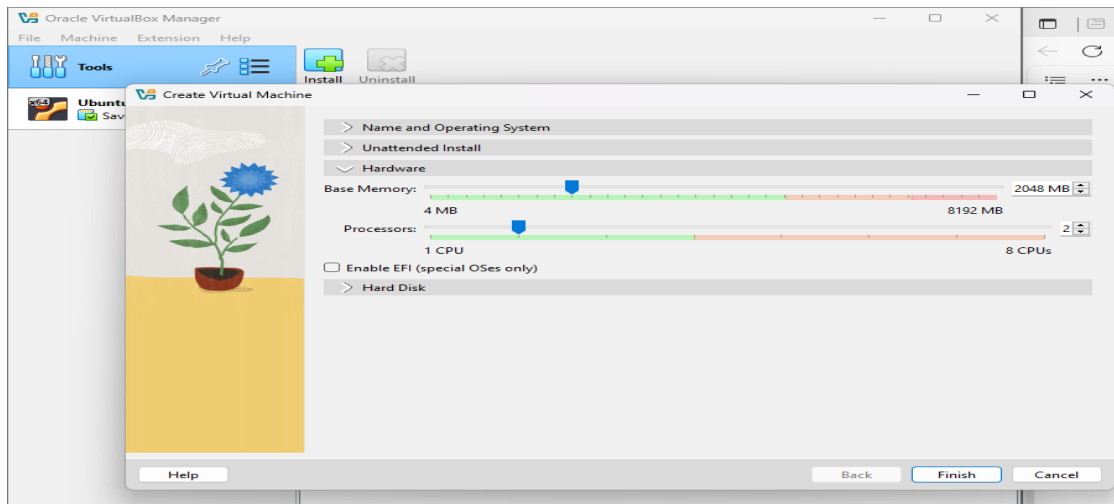


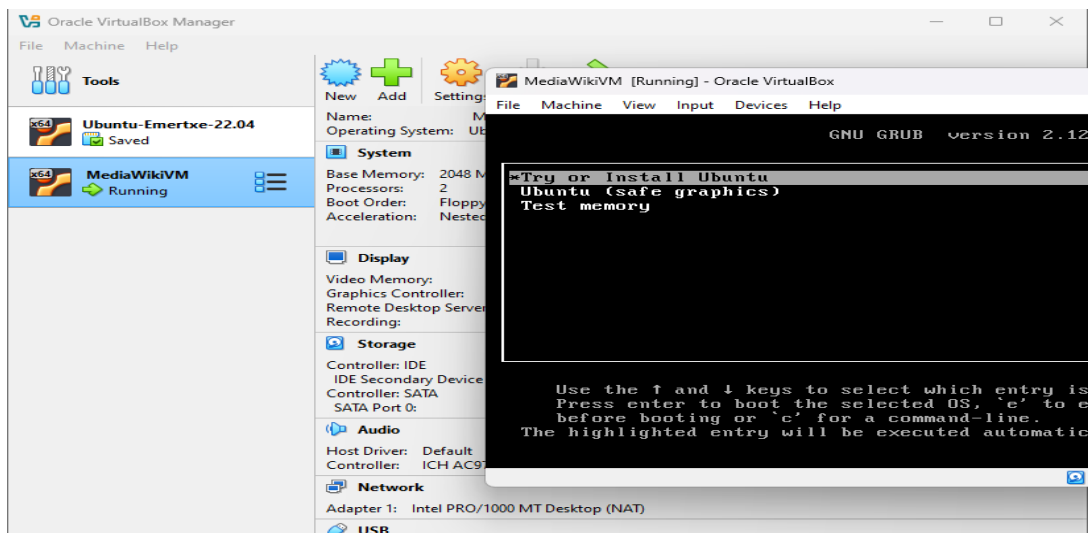
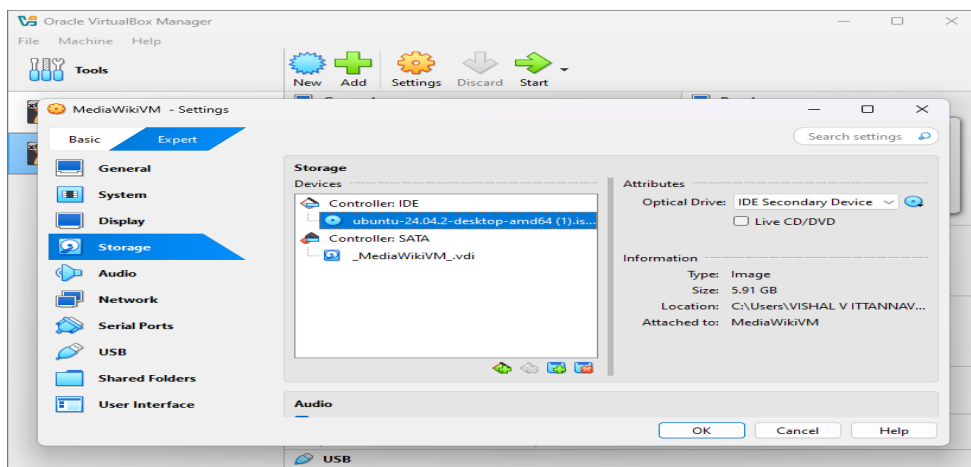
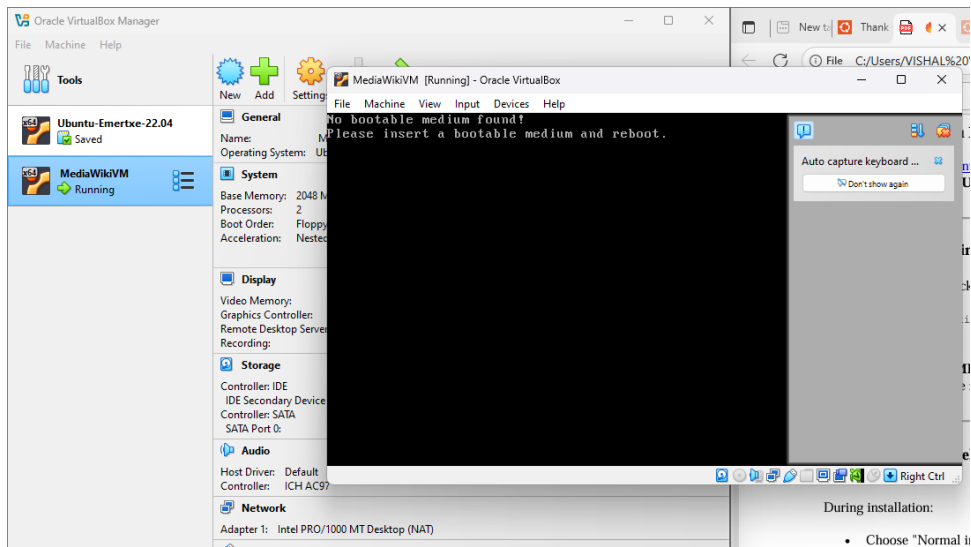
Then next process

Open VirtualBox → Click “New” → Set:

- Name: MediaWikiVM • Type: Linux
- Version: Ubuntu (64-bit)
- Memory: 2048 MB
- Hard disk: Create new VDI (20 GB+)

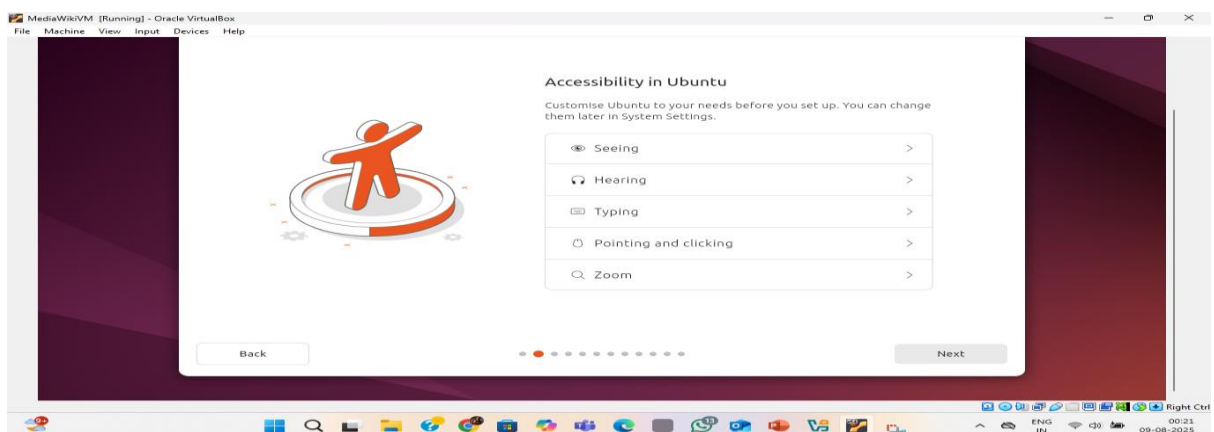
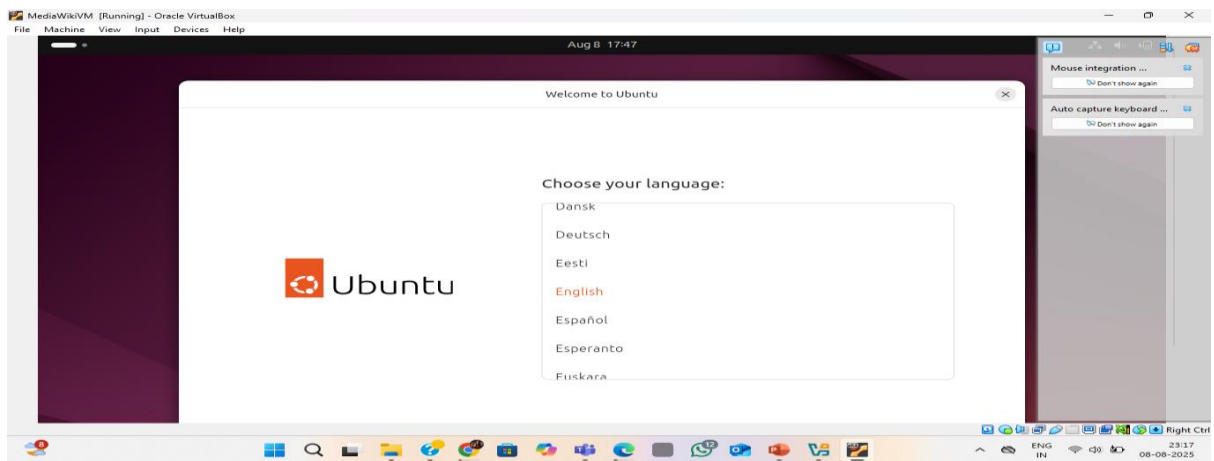
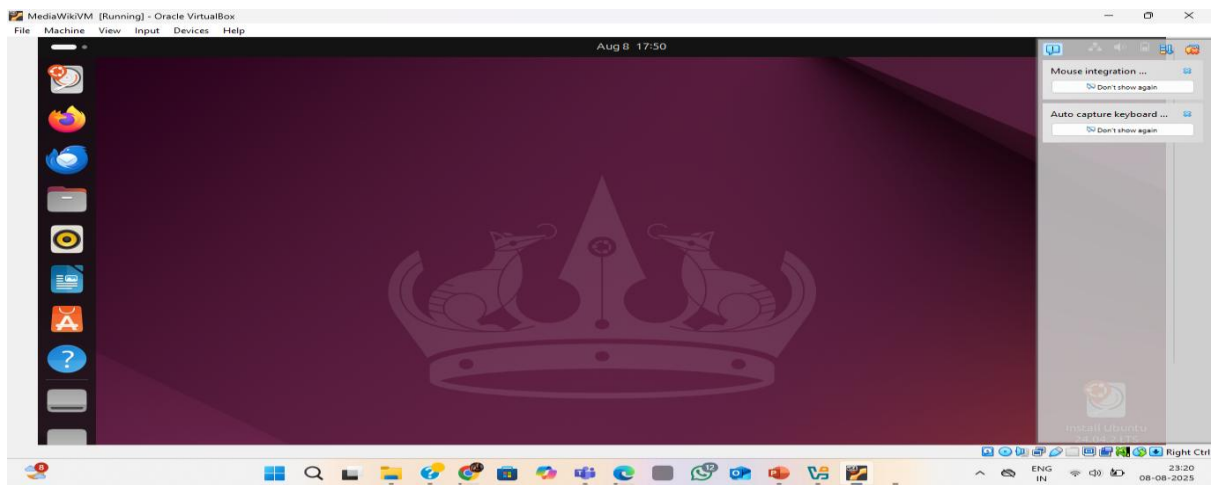


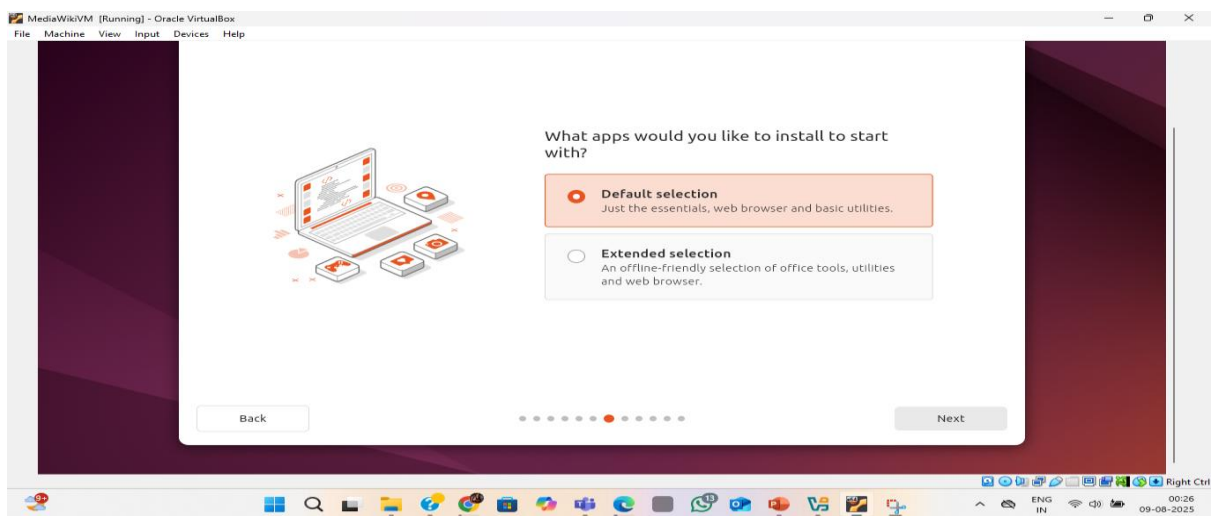
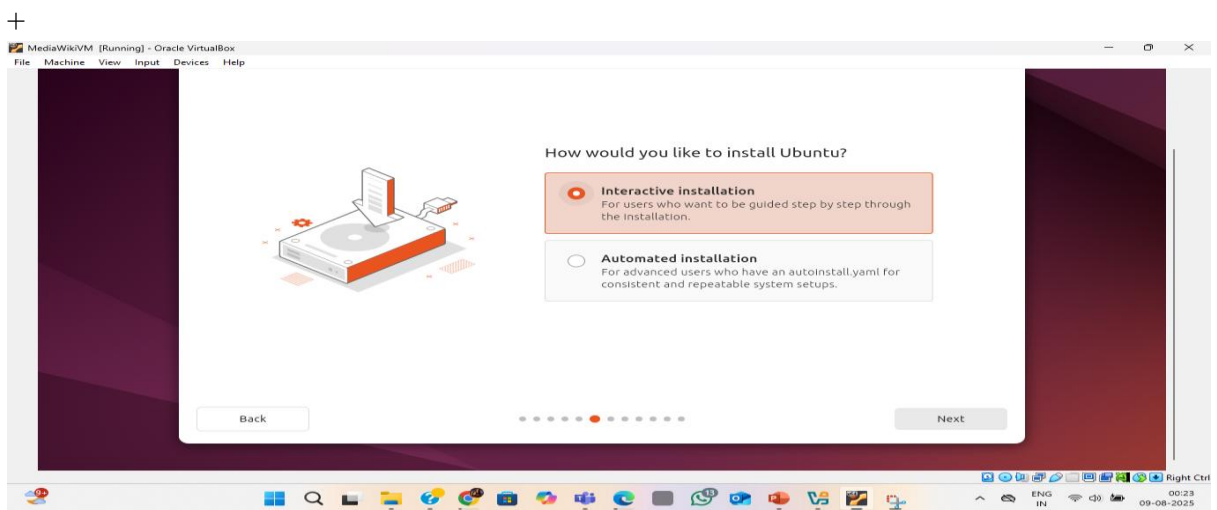
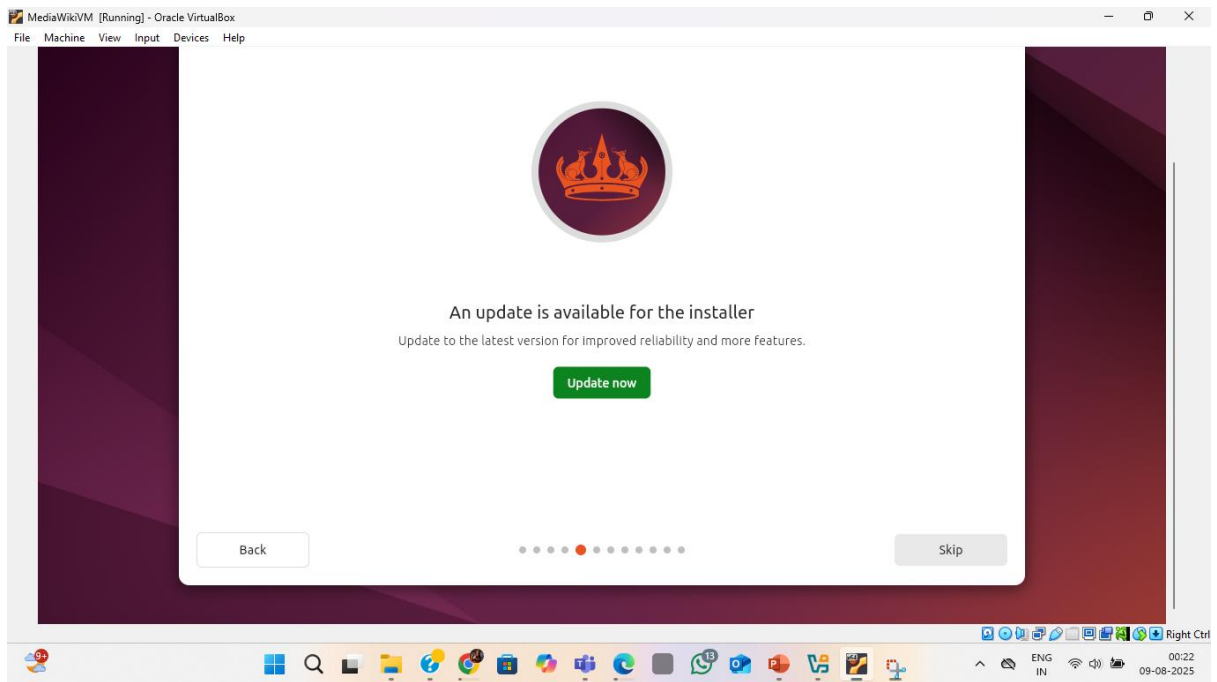


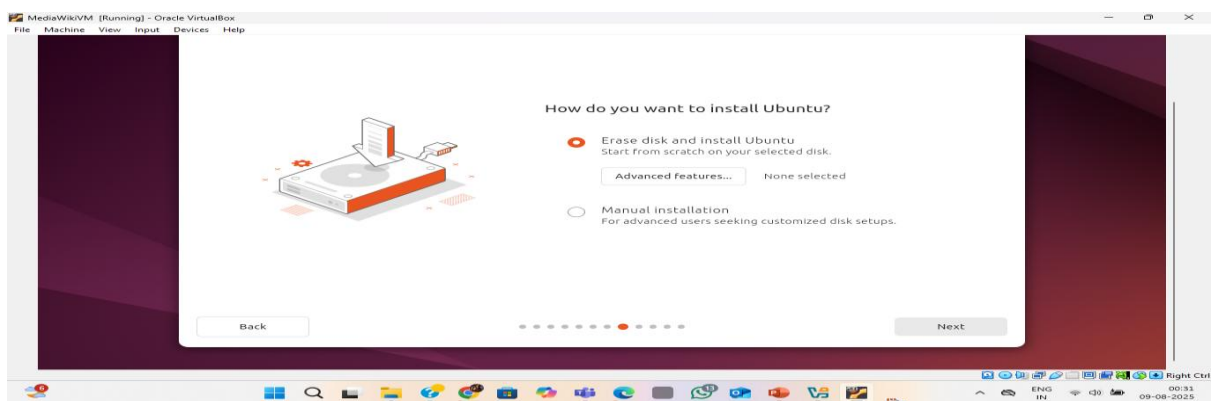
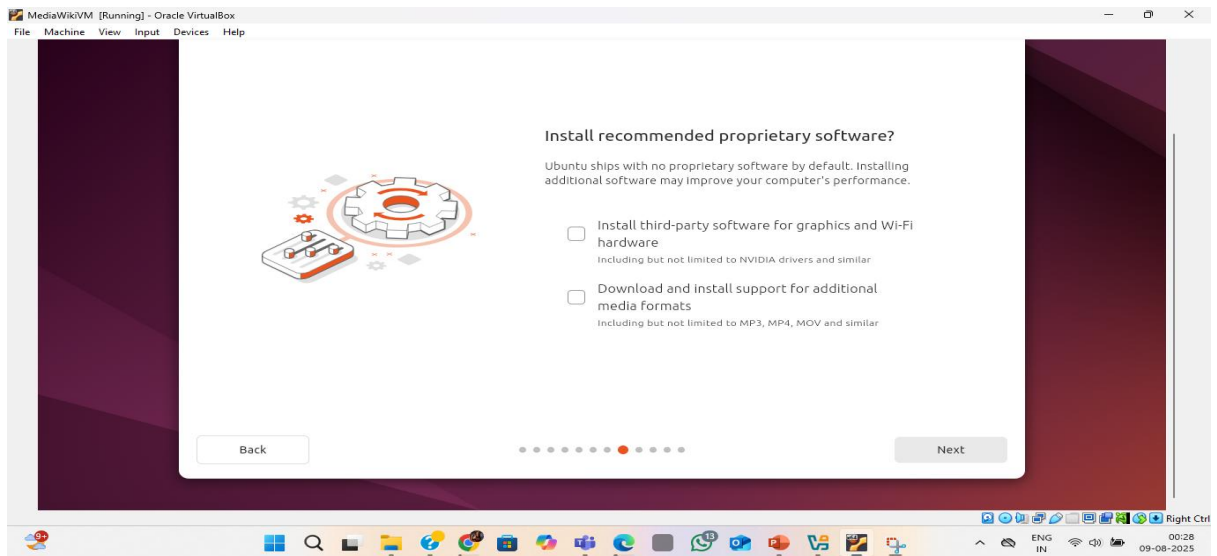


Then it will open the ubuntu

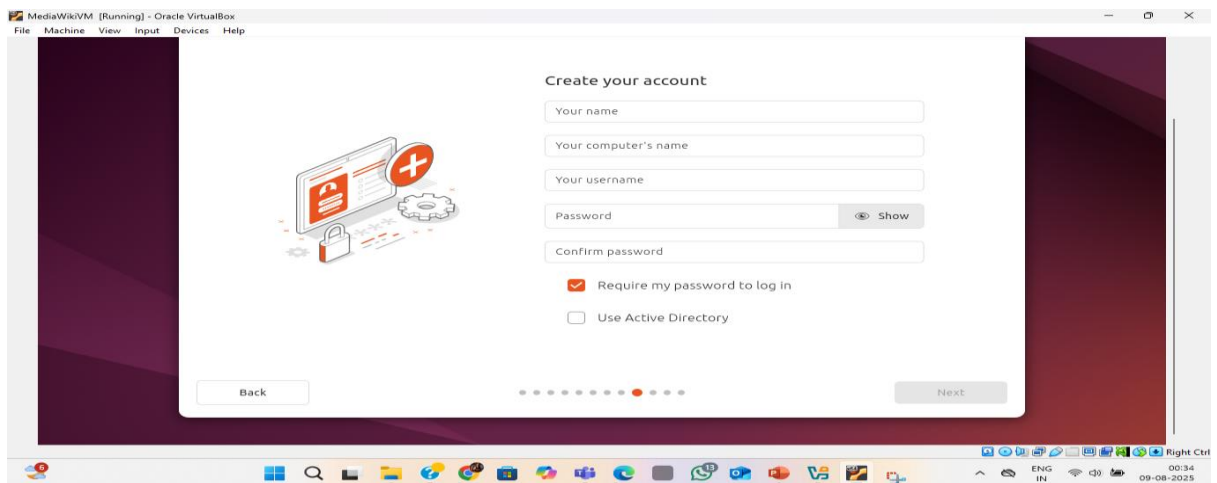
Start the VM → Select the downloaded Ubuntu ISO → Install Ubuntu inside the VM







S



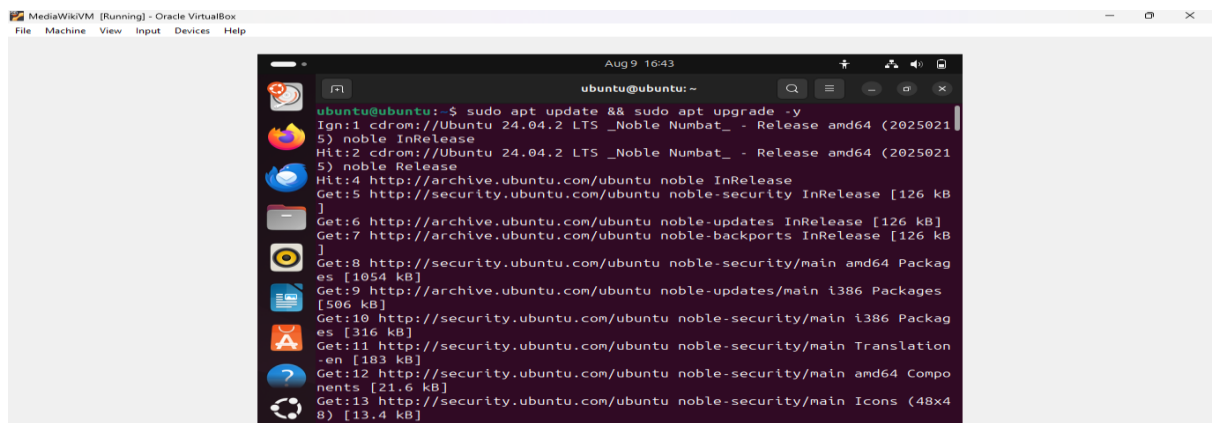
5.2 : Install the LAMP Stack (Apache + MySQL + PHP)

Open the Terminal inside Ubuntu VM

1. Update the system bash

`sudo apt update && sudo apt upgrade -y`

#This updates all packages.



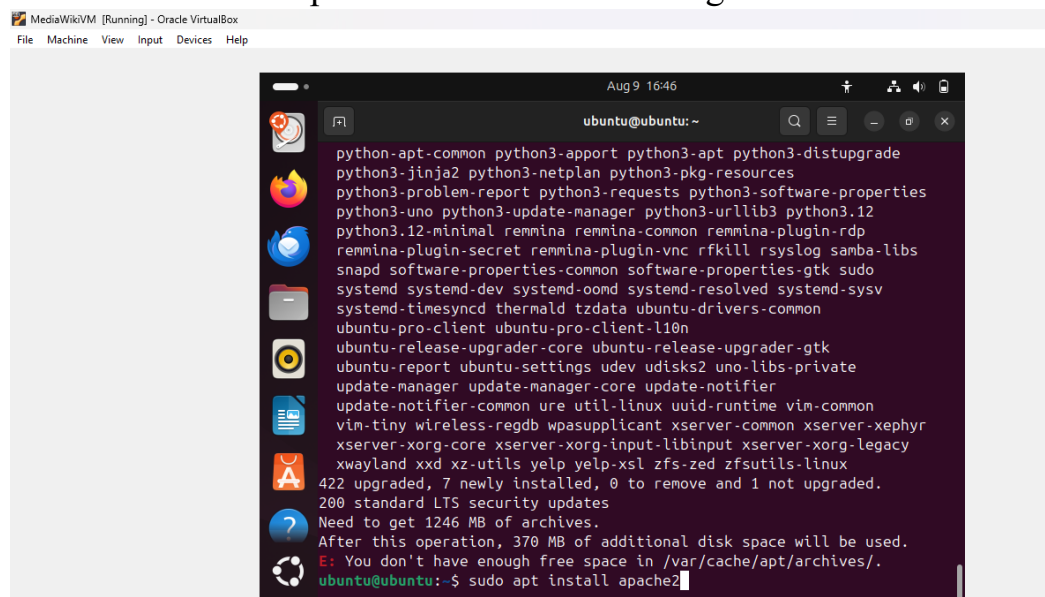
```
ubuntu@ubuntu:~$ sudo apt update && sudo apt upgrade -y
Ign:1 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble InRelease
Hit:2 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble Release
Get:3 http://archive.ubuntu.com/ubuntu noble InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1054 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main i386 Packages [506 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [316 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [183 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/main Icons [48x48] [13.4 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/main Icons [64x64] [13.4 kB]
```

2. Install Apache (web server)

`sudo apt install apache2 -y` Apache serves websites.

Check if it works:

- Open browser in VM
- Go to <http://localhost>
- You should see “Apache2 Ubuntu Default Page”



```
python3-apt-common python3-apport python3-apt python3-distupgrade python3-jinja2 python3-netplan python3-pkg-resources python3-problem-report python3-requests python3-software-properties python3-uno python3-update-manager python3-urllib3 python3.12 python3.12-minimal remmina remmina-common remmina-plugin-rdp remmina-plugin-secret remmina-plugin-vnc rfidkill rsyslog samba-libs snapd software-properties-common software-properties-gtk sudo systemd systemd-dev systemd-oom systemd-resolved systemd-sysv systemd-timesyncd thermald tzdata ubuntu-drivers-common ubuntu-pro-client ubuntu-pro-client-l10n ubuntu-release-upgrader-core ubuntu-release-upgrader-gtk ubuntu-report ubuntu-settings udev udisks2 uno-libs-private update-manager update-manager-core update-notifier update-notifier-common ure util-linux uuid-runtime vim-common vim-tiny wireless-regdb wpasupplicant xserver-common xserver-xephyr xserver-xorg-core xserver-xorg-input-libinput xserver-xorg-legacy xwayland xxd xz-utils yelp yelp-xsl zfs-zed zfsutils-linux
422 upgraded, 7 newly installed, 0 to remove and 1 not upgraded.
200 standard LTS security updates.
Need to get 1246 MB of archives.
After this operation, 370 MB of additional disk space will be used.
E: You don't have enough free space in /var/cache/apt/archives/.
ubuntu@ubuntu:~$ sudo apt install apache2
```

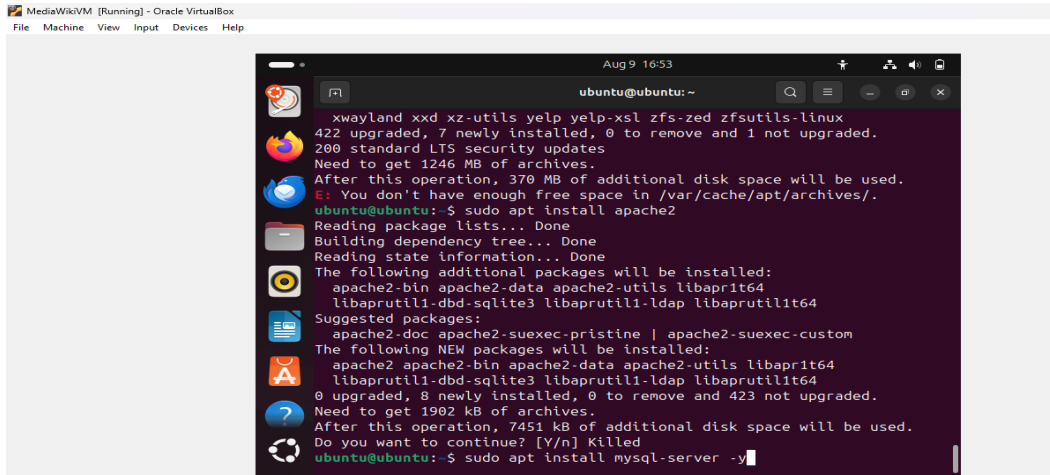

3. Install MySQL (database)

`sudo apt install mysql-server -y MySQL`

stores your wiki content.

Set root password:

`sudo mysql_secure_installation` → Follow prompts (you can skip password validation if confused).



```
MediaWikiVM [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Aug 9 16:53
ubuntu@ubuntu: ~
xwayland xxd xz-utils yelp yelp-xsl zfs-zed zfsutils-linux
422 upgraded, 7 newly installed, 0 to remove and 1 not upgraded.
Need to get 1246 MB of archives.
After this operation, 370 MB of additional disk space will be used.
E: You don't have enough free space in /var/cache/apt/archives/.
ubuntu@ubuntu:~$ sudo apt install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64
  libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64
  libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
0 upgraded, 8 newly installed, 0 to remove and 423 not upgraded.
Need to get 1902 kB of archives.
After this operation, 7451 kB of additional disk space will be used.
Do you want to continue? [Y/n] Killed
ubuntu@ubuntu:~$ sudo apt install mysql-server -y
```

4. a database for MediaWiki Type:

`sudo mysql -u root -p` Enter password.

Then type: `sql CopyEdit CREATE DATABASE wikidb;`

`CREATE USER 'wikiuser'@'localhost' IDENTIFIED BY 'wikisecret';`

`GRANT ALL PRIVILEGES ON wikidb.* TO 'wikiuser'@'localhost';`

`FLUSH PRIVILEGES;`

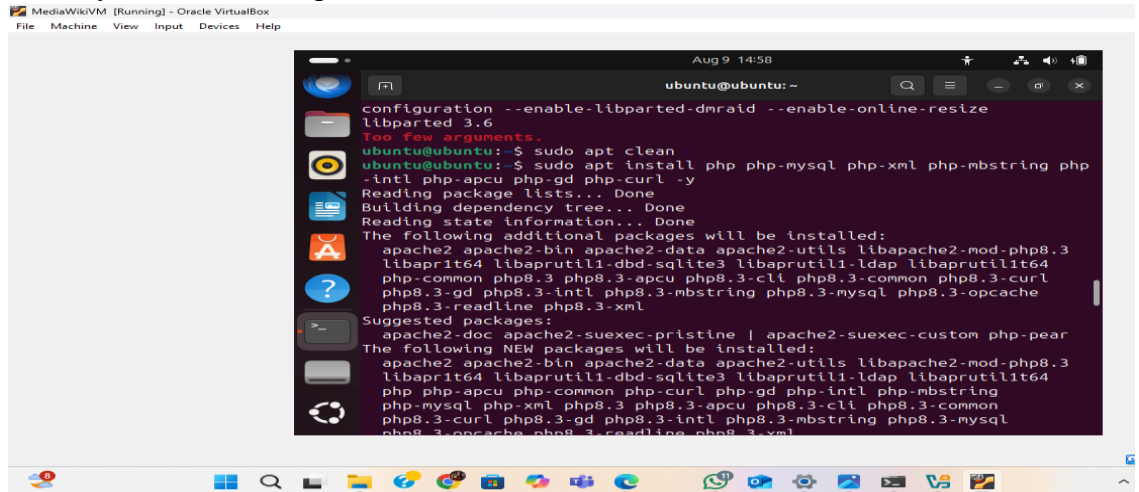
`EXIT;`

5. Install PHP and required extensions

```
sudo apt install php php-mysql php-xml php-mbstring php-intl php-apcu php-gd php-curl -y
```

Restart Apache:

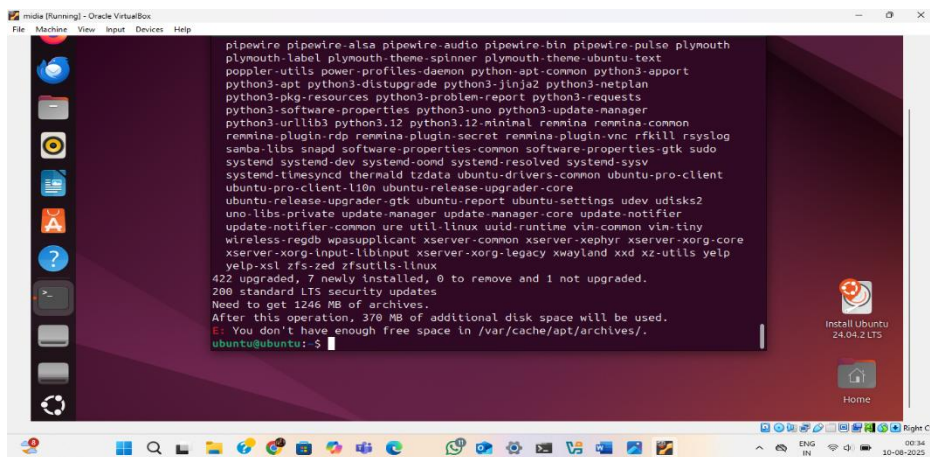
```
sudo systemctl restart apache2
```



The screenshot shows a terminal window titled 'ubuntu@ubuntu: ~' with a search bar and window controls. The terminal output shows the following commands and their results:

```
configuration --enable-libparted-dmraid --enable-online-resize
libparted 3.6
Too few arguments.
ubuntu@ubuntu:~$ sudo apt clean
ubuntu@ubuntu:~$ sudo apt install php php-mysql php-xml php-mbstring php-intl php-apcu php-gd php-curl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapache2-mod-php8.3
  libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  php-common php8.3 php8.3-apcu php8.3-cli php8.3-common php8.3-curl
  php8.3-gd php8.3-intl php8.3-mbstring php8.3-mysql php8.3-opcache
  php8.3-readline php8.3-xml
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom php-pear
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapache2-mod-php8.3
  libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  php php-apcu php-common php-curl php-gd php-intl php-mbstring
  php-mysql php-xml php8.3 php8.3-apcu php8.3-cli php8.3-common
  php8.3-curl php8.3-gd php8.3-intl php8.3-mbstring php8.3-mysql
  php8.3-opcache php8.3-readline php8.3-xml
```

After this step I got many error and problems



5.3 Problem Faced with LAMP Setup

During the installation of the LAMP stack on Ubuntu Server, the virtual machine experienced network configuration issues due to the NAT network mode in VirtualBox, preventing external access to the MediaWiki site from the host machine.

Additionally, the installation process required multiple Linux-specific commands for Apache, MySQL, and PHP configuration, which increased setup complexity and led to dependency errors when installing PHP extensions.

These issues delayed progress and made it impractical to complete the setup within the given time frame.

As a result, the decision was made to switch to WAMP on Windows, which offers a pre-configured environment and faster deployment for testing and demonstration purposes.

6.Reason for Using WAMP Instead of LAMP

While the original plan was to set up MediaWiki on a Linux VM using a LAMP stack, the setup could not be completed due to **technical and compatibility issues**:

Problems Faced with LAMP Setup:

1. **VM Performance Issues** – The VirtualBox VM was running very slowly due to limited available RAM and disk space on the host system.
2. **Network Configuration Problems** – Apache on the VM was not reachable from the host machine due to NAT/bridged network misconfiguration.
3. **Package Installation Errors** – Certain PHP and MySQL dependencies failed to install correctly because of outdated package mirrors and network timeouts

Time Constraints – Troubleshooting the above issues exceeded the available time, so an alternative stack was required.

Decision:

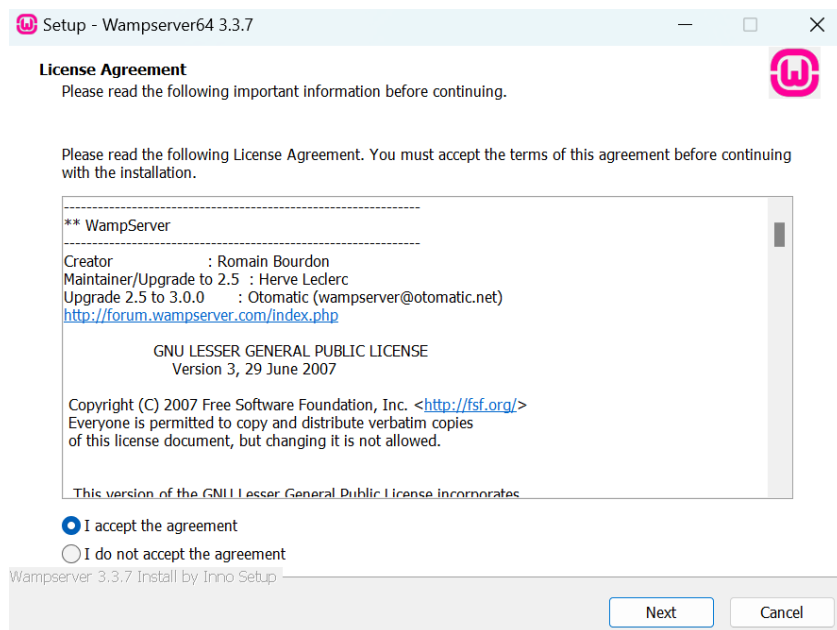
Switched to **WAMP on Windows** to ensure faster setup, easier debugging, and completion of the MediaWiki installation within the given timeframe.

7.1 Installing WAMP Server

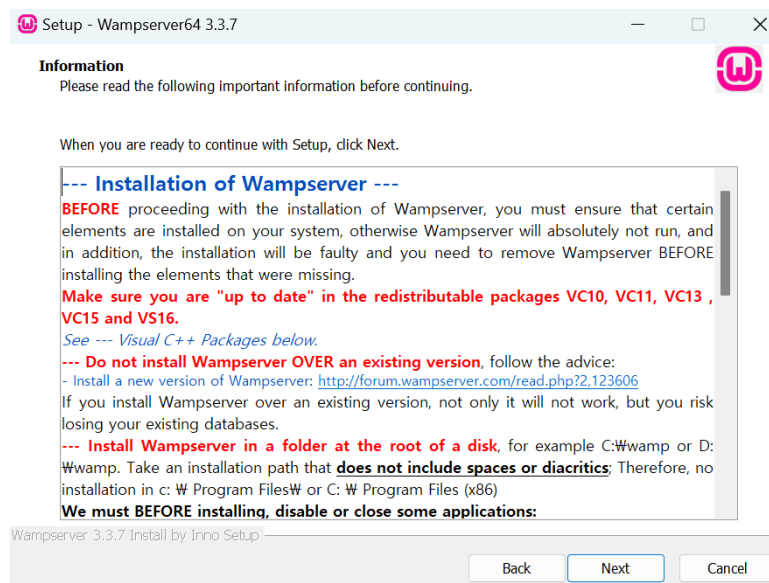
1. Download WAMP from the official site: <https://www.wampserver.com>.
2. Run the installer and choose:
 - **Installation Directory:** C:\wamp64
 - **Default Browser:** Choose your preferred browser.
3. Once installed, start WAMP and ensure the **icon in the taskbar turns green** (meaning Apache & MySQL are running).

STEPS:

Click I accept the agreement



CLICK NEXT:



If you encounter the following error:



A. For WAMP 3.3.x and above, you typically need:

- VC++ 2008 (x86 + x64)
- VC++ 2010 (x86 + x64)
- VC++ 2012 (x86 + x64)
- VC++ 2013 (x86 + x64)
- VC++ 2015-2022 (x86 + x64) – this covers 2015, 2017, 2019, 2022 in one package

B. Download the Required VC++ Packages

WAMP has an official help page listing the required downloads:

 **WAMP VC++ Packages Download Page-**

<https://learn.microsoft.com/en-us/cpp/windows/latest-supported-vc-redist?view=msvc-170>

Download and install:

1. **VC++ 2008 Redistributable – x86 and x64**
2. **VC++ 2010 Redistributable – x86 and x64**
3. **VC++ 2012 Redistributable – x86 and x64**
4. **VC++ 2013 Redistributable – x86 and x64**
5. **VC++ 2015-2022 Redistributable – x86 and x64** (latest version from Microsoft: Download link and x86 version)

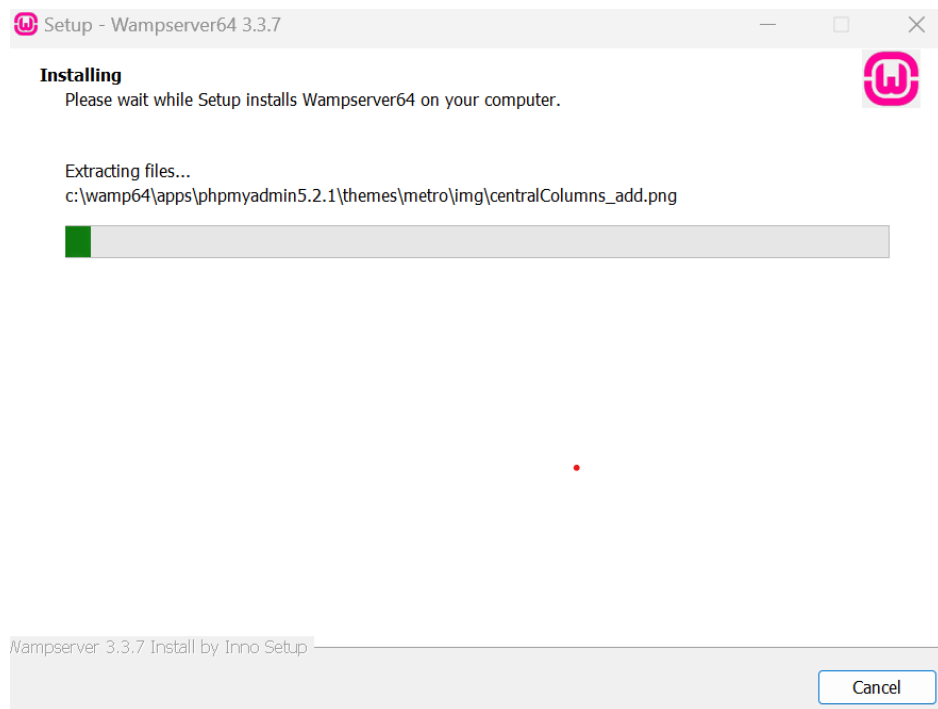


C. Installation Order

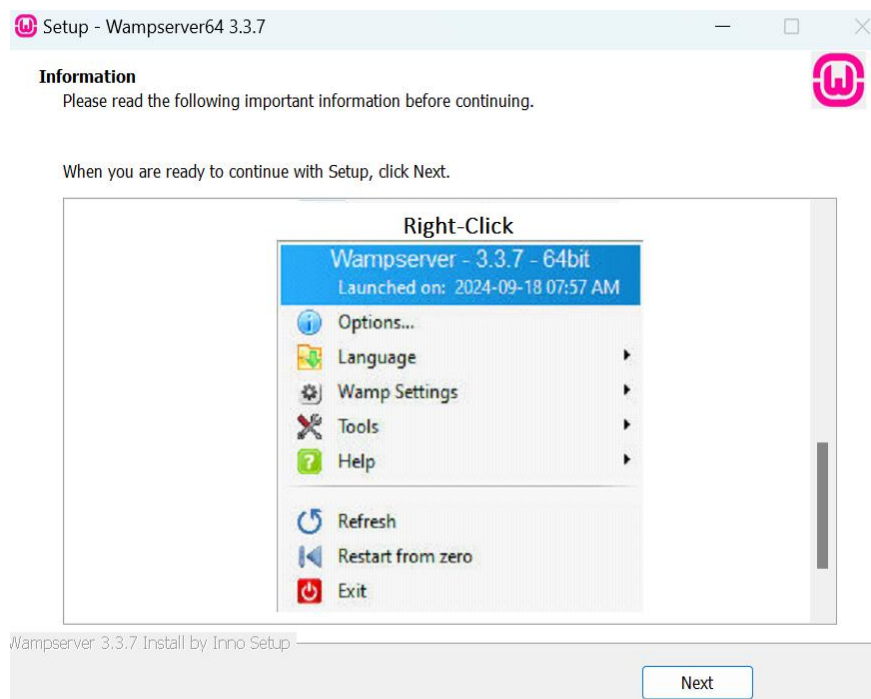
To avoid conflicts:

1. Install all **x86** versions first.
2. Install all **x64** versions second.
3. Restart your computer.

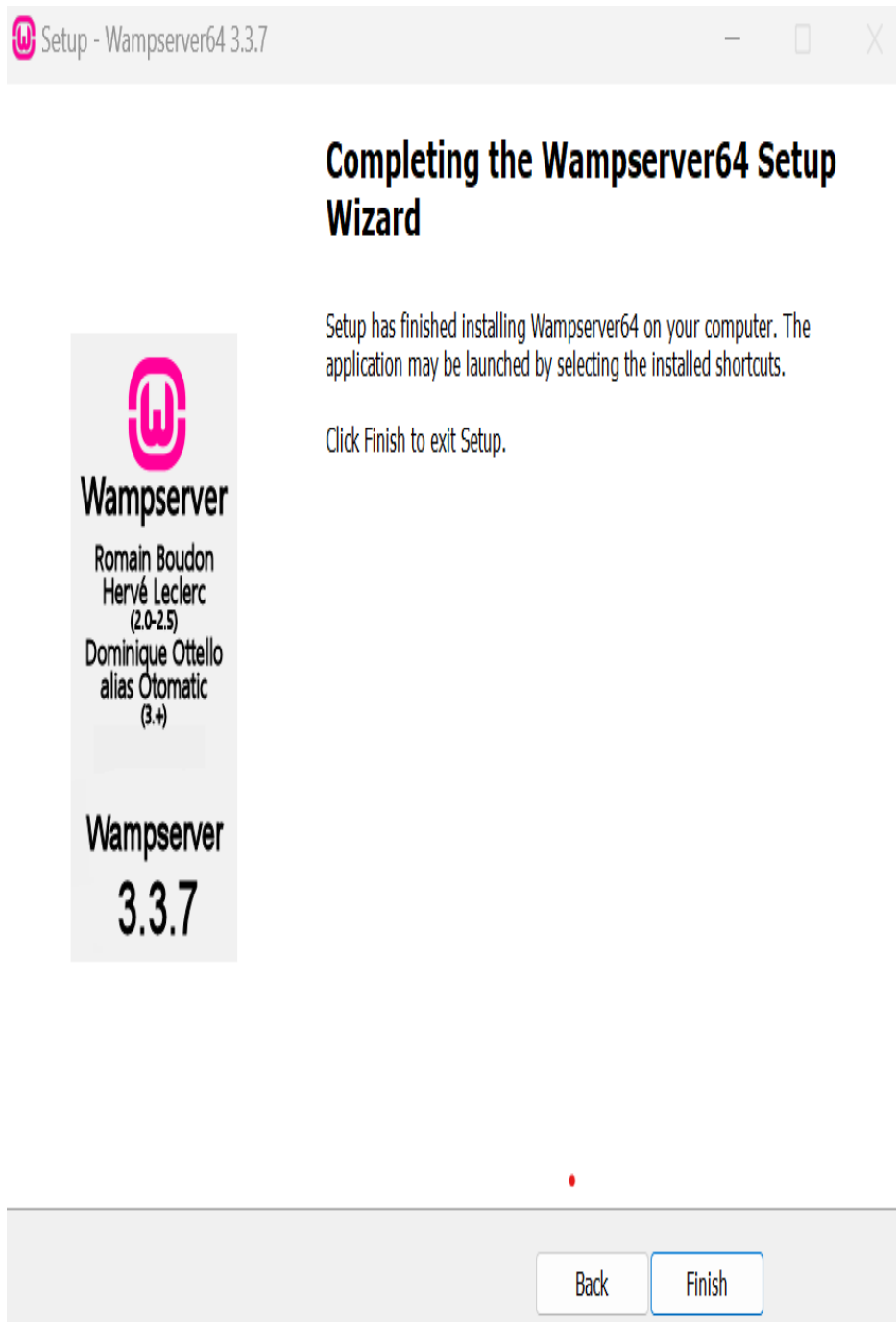
AFTER INSTALLING ALL PACKAGE INSTALLL WAMP:



CLICK NEXT,NEXT:



CLICK FINISH:



7.2 Downloading and Placing MediaWiki

Prerequisites

- WAMP is installed and running (tray icon → green ideally).
- You have admin (or at least elevated) rights on Windows.
- An unzip tool (Windows Explorer works; 7-Zip recommended).
- Browser for the installer (Chrome/Edge/Firefox).

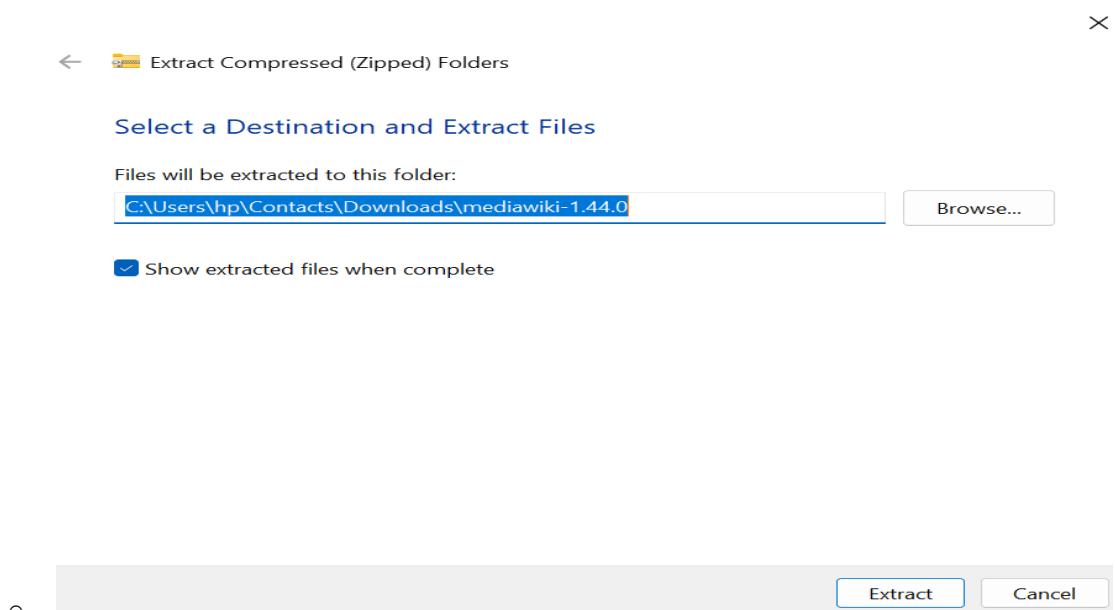
A. GUI (easy) — recommended for most users

1. Download MediaWiki

- Open: <https://www.mediawiki.org/wiki/Download>
- Click the stable release **ZIP** link and save the file (usually to C:\Users\<you>\Downloads).

2. Extract the ZIP

- Open File Explorer → go to Downloads.
- Right-click the mediawiki-<version>.zip → **Extract All...** → extract to Downloads (this creates a folder like mediawiki-1.42.0).

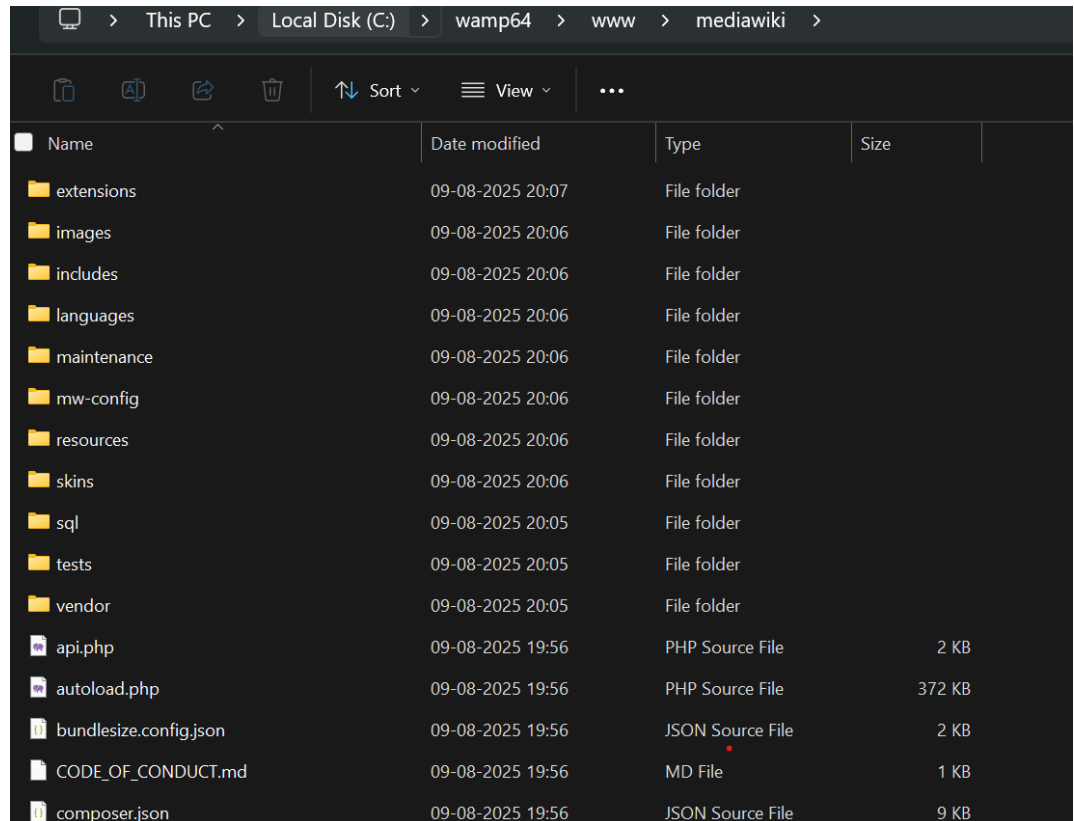


3. Rename the extracted folder (optional but tidy)

- Right-click the folder mediawiki-1.42.0 → **Rename** → mediawiki.
-

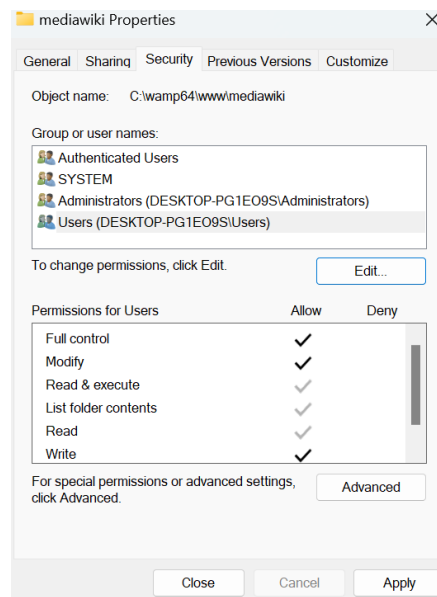
4. Move the folder into WAMP's www

- Open another File Explorer window and go to C:\wamp64\www.
- Drag or Cut + Paste the folder mediawiki into C:\wamp64\www.



5. Set basic write permissions for uploads (images)

- Right-click C:\wamp64\www\mediawiki → **Properties** → **Security** tab → **Edit**.
- Select **Users** (or add Everyone for a temporary dev setup) → check **Modify** (gives write).
- Click **Apply** → **OK**.



- For safer minimal change, only give **Modify** to the images folder (see Troubleshooting step).

6. Restart WAMP

- Left-click the WAMP tray icon → **Restart All Services**.

7. Open the MediaWiki web installer

- In your browser go to: <http://localhost/mediawiki>
- Follow the on-screen installation wizard (DB details, admin credentials, site name).
- At the end the installer provides a LocalSettings.php file to download — save it.

7.3 FOLLOW FOLLOWING STEPS DATABASE SETUP(MYSQL):

After placing MediaWiki in C:\wamp64\www\mediawiki, you open your browser and go to:

<http://localhost/mediawiki>

IMG-7.1 CLICK SETUP



MediaWiki 1.44.0

LocalSettings.php not found.

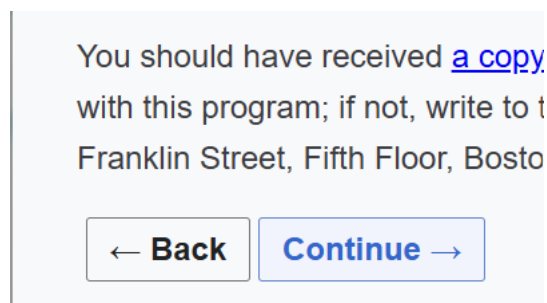
Please [set up the wiki](#) first.

IMG-7.2 SELECT LANGUAGE, CLICK NEXT



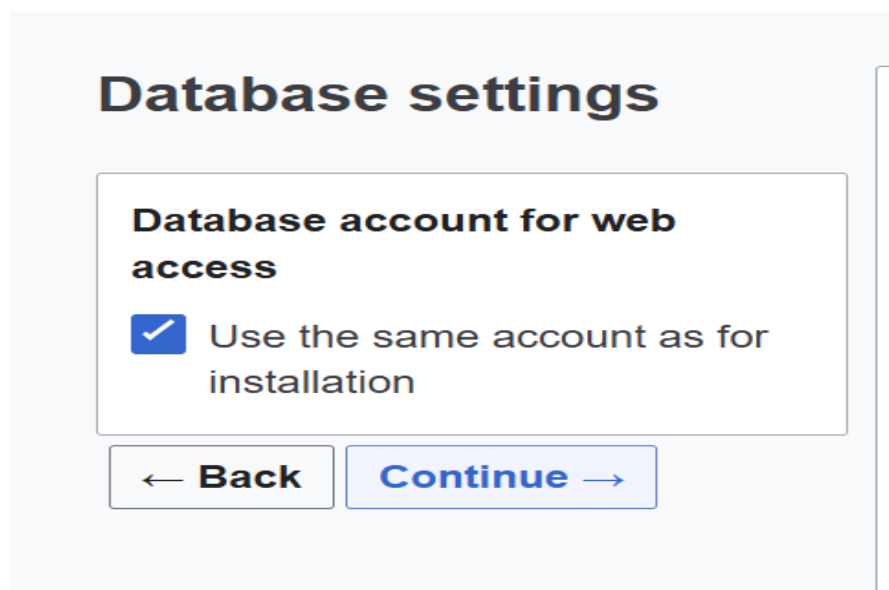
A screenshot of a language selection interface. At the top right, there is a link that says "Restart in:". Below this, there is a dropdown menu showing "en-gb - British English" with a downward arrow. Underneath the dropdown, the text "Wiki language:" is followed by a blue information icon and the word "help". Below that is another dropdown menu, also showing "en-gb - British English" with a downward arrow. At the bottom of the interface is a blue button with the text "Continue" and a right-pointing arrow.

IMG-7.3 AFTER ENVIRONMENT CHECK, CLICK NEXT



A screenshot of a message box. The text inside reads: "You should have received [a copy](#) with this program; if not, write to t Franklin Street, Fifth Floor, Bosto". At the bottom of the box are two buttons: a "Back" button with a left-pointing arrow and a "Continue" button with a right-pointing arrow.

IMG-7.4 CLICK NEXT:



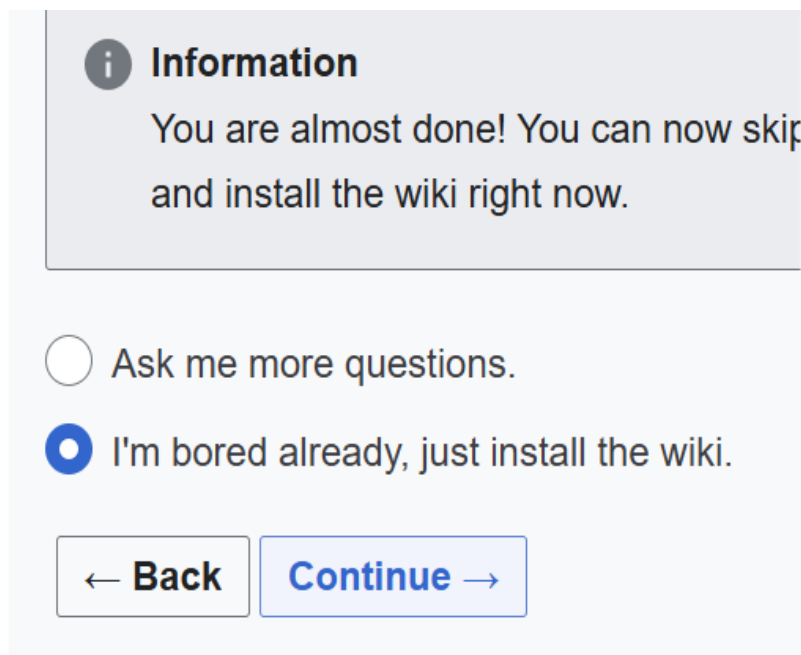
Database settings

Database account for web access

☒ Use the same account as for installation

← **Back** **Continue** →

IMG-7.5 SET ROOT NAME, PASSWORD , CLICK NEXT, CLICK NEXT AGAIN:



i Information

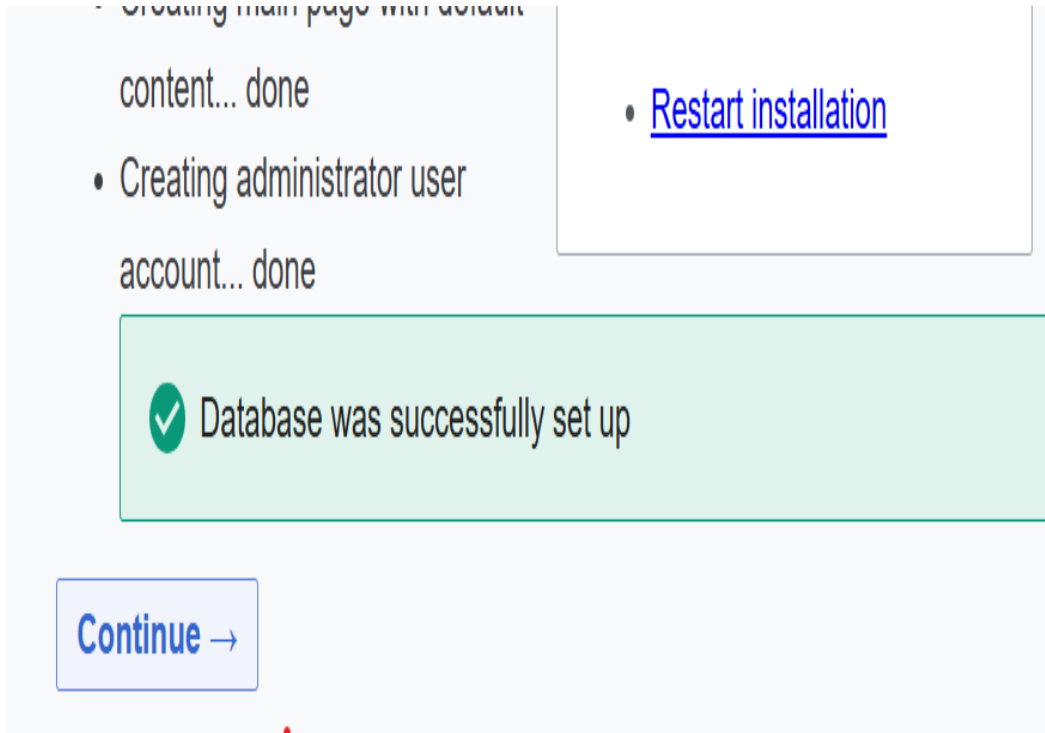
You are almost done! You can now skip and install the wiki right now.

☐ Ask me more questions.

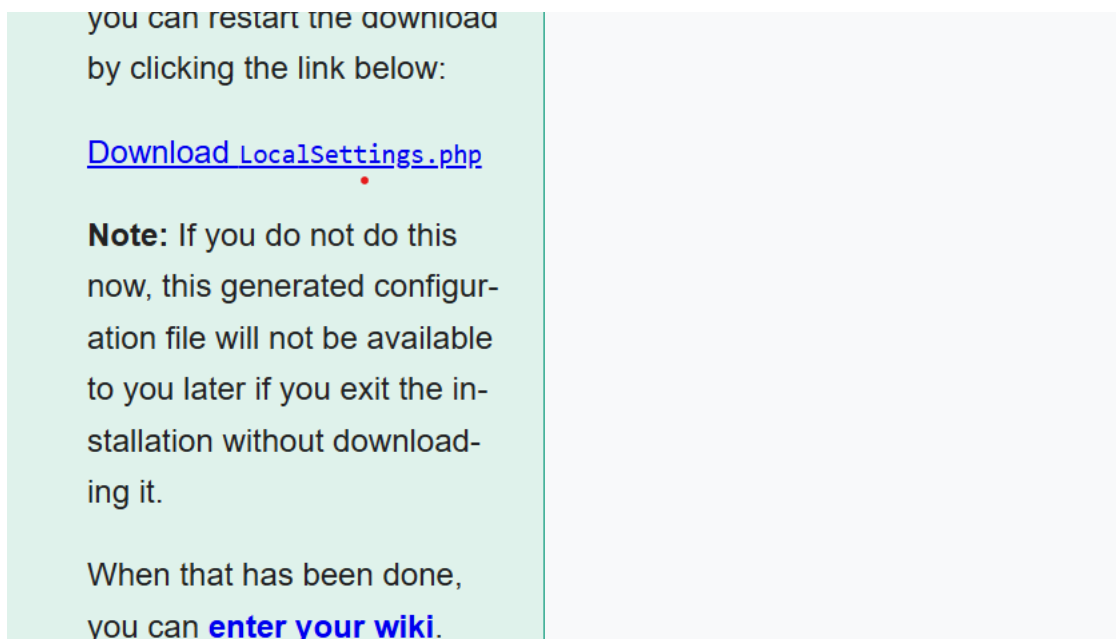
☒ I'm bored already, just install the wiki.

← **Back** **Continue** →

IMG-7.5 DATABASE SETUP COMPLETED:



DOWNLOAD LOCALSETTINGS.PHP FILE:



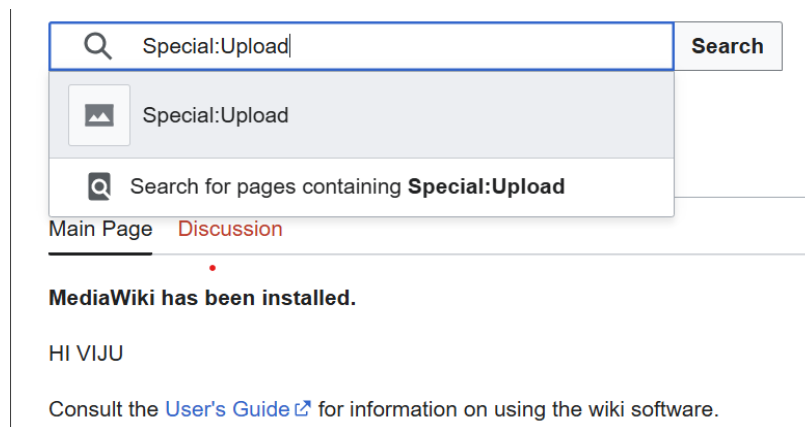
8. Place LocalSettings.php

- Copy the downloaded LocalSettings.php into C:\wamp64\www\mediawiki\ (overwrite if prompted).

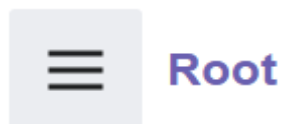
9. Verify

- Visit <http://localhost/mediawiki> — your wiki should load.
- Create a test page and upload an image to confirm images/ is writable.

IN SEARCH BOX SEARCH-SPECIAL:UPLOAD



If uploads disabled like the below, follow the below steps:



Uploads disabled.

File uploads are disabled.

Return to [Main Page](#).

- Open C:\wamp64\www\mediawiki\LocalSettings.php
- Add or modify:

\$wgEnableUploads = true;

After it if it ask for password and username, log in using your login creditionals

Upload image look like these:



Root

File:Thar.jpg

[Main menu](#)
[hide](#)

[Main page](#)
[Recent changes](#)
[Random page](#)
[Help about MediaWiki](#)
[Special pages](#)
[Switch to old look](#)

[File](#)
[Discussion](#)

[File](#)
[File history](#)
[File use](#)



No higher resolution available.

[Thar.jpg](#) (275 × 183 pixels, file size: 10 KB, MIME type: image/jpeg)

8. Set Up the Backup Mechanism.

8.1 Step 1 — Find your exact mysqldump.exe path

1. Open C:\wamp64\bin\mysql\ in File Explorer.
2. Inside, you'll see a folder like mysql8.0.36 (the exact number will vary).
3. Inside that → bin → you should see mysqldump.exe.
4. Copy that full path, e.g.:

python

C:\wamp64\bin\mysql\mysql9.1.0\bin\mysqldump.exe

8.2 Step 2 — Create the backup folder

1. Press **Win+E** → go to C:\
2. Right click → **New** → **Folder** → name it:

nginx

wiki_backup

3. You'll now have C:\wiki_backup

8.3 Step 3 — Create the PowerShell script:

a. Press Win+S, type notepad, right-click → Run as administrator

b. Paste this script (edit DB name and MySQL path as needed):

```
# === backup_all.ps1 ===
```

```
$backupRoot = "C:\wiki_backup"
```

```
$dbName    = "wikidb"           # change if your DB name differs
```

```
$dbUser     = "root"
```

```
$dbPass     = "YourDBPassword"  # replace with actual password
```

```

$mwPath    = "C:\wamp64\www\mediawiki"

$keepDays  = 14

$date = Get-Date -Format "yyyy-MM-dd_HH-mm-ss"

$backupDir = Join-Path $backupRoot $date

New-Item -ItemType Directory -Path $backupDir -Force | Out-Null

$mysqldump = "C:\wamp64\bin\mysql\mysql8.0.36\bin\mysqldump.exe" #
update version number if needed

# DB dump

$dumpFile = Join-Path $backupDir "$dbName.sql"

& $mysqldump --user=$dbUser --password=$dbPass --single-transaction --
routines --events --triggers --databases $dbName > $dumpFile

# Zip DB dump

Compress-Archive -LiteralPath $dumpFile -DestinationPath (Join-Path
$backupDir "$dbName.zip") -Force

Remove-Item $dumpFile

# Zip MediaWiki files

Compress-Archive -Path "$mwPath\*" -DestinationPath (Join-Path $backupDir
"mediawiki_files_$date.zip") -Force

# Rotate old backups

Get-ChildItem -Path $backupRoot -Directory | Sort-Object CreationTime -
Descending | Select-Object -Skip $keepDays | Remove-Item -Recurse -Force

# Log

"$([Get-Date]) Backup completed: $backupDir" | Out-File -FilePath (Join-Path
$backupRoot "backup.log") -Append

```

C. Save it as:backup_all.ps1

path:

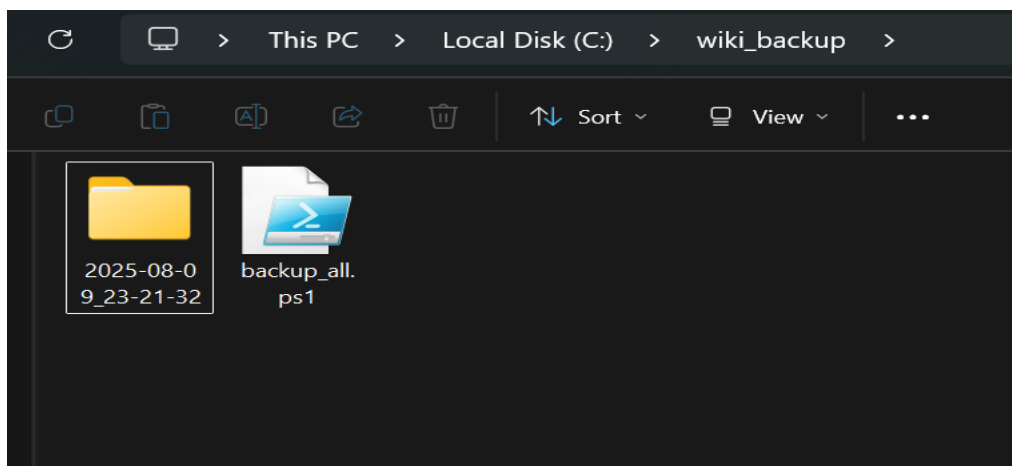
C:\wiki_backup\backup_all.ps1

8.4 Step 4 — Test the script

1. Press **Win+X** → **Windows Terminal (Admin)**
2. Run:

```
Set-ExecutionPolicy Bypass -Scope Process -Force  
powershell.exe -ExecutionPolicy Bypass -File  
"C:\wiki_backup\backup_all.ps1"
```

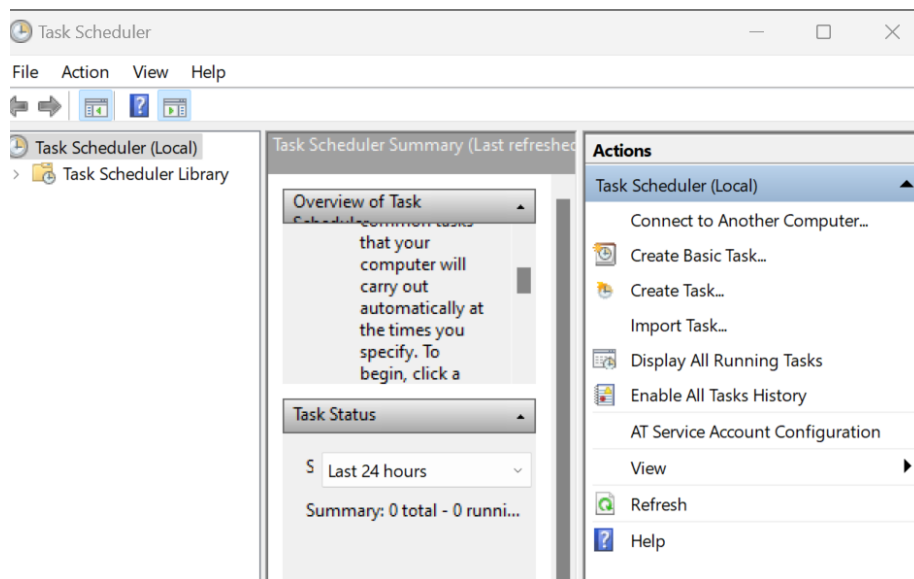
3. Check C:\wiki_backup — you should see:
 - a folder with today's date
 - inside: wikidb.zip and mediawiki_files_<date>.zip
 - a backup.log file



8.5 Step 5 — Schedule it daily in Task Scheduler

Tell Windows to run your backup script every day automatically

1. Press **Win+S**, type **Task Scheduler**, open it.



2. **Create Task** (not “basic”).
3. **General** tab:
 - Name: MediaWiki Backup
 - Check **Run whether user is logged on or not**
 - Check **Run with highest privileges**
4. **Triggers** tab → New:
 - Daily, recur every 1 day, start at 02:00 AM
5. **Actions** tab → New:
 - Program/script: powershell.exe
 - Arguments:

arduino

-ExecutionPolicy Bypass -File "C:\wiki_backup\backup_all.ps1"
6. Click OK → Enter your Windows account password

8.6 Step 6 — Test the restore

Test that backups actually work

- **Database:**

```
C:\wamp64\bin\mysql\mysql8.0.36\bin\mysql.exe -u root -p wikidb <  
C:\wiki_backup\<backup-folder>\wikidb.sql
```

(Unzip the .zip first.)

- **Files:** extract mediawiki_files_<date>.zip into
C:\wamp64\www\mediawiki.

9. Future Enhancements

To improve the setup and ensure better scalability, performance, and security, the following enhancements are recommended:

1. Complete LAMP Deployment

- Allocate higher RAM and CPU to the VM for smoother operation. Use bridged networking or proper NAT port forwarding for external access. Resolve package mirror and dependency issues for a successful Ubuntu-based setup.

2. SSL/TLS Configuration

- Secure the MediaWiki site with HTTPS using **Let's Encrypt** or a custom SSL certificate.

3. Offsite & Cloud Backup

- Store backups in a secure offsite location or cloud storage (e.g., AWS S3, Google Drive) for disaster recovery.

4. Incremental Backups

- Implement incremental backups to reduce storage usage and backup time.

5. Monitoring & Alerts

- Set up server monitoring tools (e.g., Nagios, Zabbix, or UptimeRobot) to track performance and uptime. Configure email or SMS alerts for backup failures or downtime.

6. User Authentication Enhancements

- Integrate MediaWiki with LDAP, Active Directory, or Single Sign-On (SSO) for better user management in enterprise environments.

7. Automated Deployment Scripts

- Create shell or PowerShell scripts for automated installation and configuration to speed up deployment in future projects.

8. Performance Optimization

- Enable caching (e.g., Memcached, APCu) for faster page loads. Optimize MySQL database queries and indexes.

9. Regular Security Updates

Apply security patches to Apache, PHP, MySQL, and MediaWiki regularly. Enable a firewall and harden the server against common web attacks.

10. Conclusion

This project successfully demonstrated the deployment and configuration of **MediaWiki** on a virtualized server environment using both **LAMP (Linux, Apache, MySQL, PHP)** and **WAMP (Windows, Apache, MySQL, PHP)** stacks.

Although the initial goal was to complete the setup on a LAMP stack in Ubuntu, technical limitations such as **network misconfiguration, package dependency issues, and VM performance constraints** necessitated a switch to the WAMP stack.

The **WAMP-based deployment** enabled a faster setup process with easier configuration and troubleshooting, ensuring the MediaWiki installation was completed within the required timeframe.

A **daily automated backup mechanism** was implemented to secure both the **MySQL database** and **MediaWiki files**, providing a reliable recovery option in the event of data loss.

Testing confirmed that the backup and restore process works as expected, ensuring business continuity and data integrity.

This documentation not only serves as a technical reference for deploying MediaWiki but also as a troubleshooting guide for overcoming common setup challenges.

